COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

Ecology, Evolution and Environmental Biology

EEEBS 3015 Animal Behavior Through Fieldwork
MW 1:00pm-4:10pm
3 credits
Elective

Instructor: Malcolm Rosenthal, mfr2176@columbia.edu

Office Hours: Thursdays, 1-3PM (ZOOM), or by appointment.
Response Policy: I will generally respond to emails within 24 hours during the week. Messages sent over the weekend or late on Friday may not receive a response until the following Monday.

Course Overview

Why do birds sing? Why do wolves hunt in packs, but spiders hunt alone? Why are worker bees willing to die to protect the queen? Humans have long been fascinated with figuring out why animals do the things they do. Our goal will be to understand how ecological forces drive the evolution of animal behavior. Students will gain experience with the fundamental theoretical toolkit of behavioral ecology and evolutionary biology. Students will then use these tools to explore diverse behavioral topics from mating and parental care, to communication and social behavior, to foraging and predation.

Animal behavior stands out among the hard sciences in terms of accessibility. A notebook and a pair of good walking shoes is all the equipment needed to get started in field research. An online-only course offers the unique opportunity to share observations and experiences with animals from across a diverse spectrum of environments and habitats. This course is intended to bring students together during class time for collaborative work and lecture-based learning, as well as to support each student in exploring the environment they live in and the animals they share it with.

This course does not require any prerequisites other than an interest in animal behavior. We will cover necessary concepts in evolution and ecology, as well as strategies for reading complex scientific articles

Learning Objectives

By the end of this course, you will be able to:

1. Build a foundation in the basic theories of evolution and natural selection
2. Develop a behavioral ecology framework for understanding how animal behaviors evolve in form and function
3. Apply the principles of critical reasoning, data interpretation, and hypothesis testing
4. Get hands-on experience with field research, animal observation, and data collection
5. Produce concise, well-structured writings & presentations that translate complex ideas into clear, persuasive arguments.

Readings

Readings for this course are drawn from: **Animal Behavior, 11th Ed. by Dustin R. Rubenstein & John Alcock, Oxford University Press, Sunderland, MA1**, as well as from selected peer-reviewed research articles (pdf files will be provided on Canvas).

Peer-Reviewed Articles:

Altmann, J. (1974). Observational study of behavior: Sampling methods. *Behaviour, 49*, 227–266.

Bateson, P., & Laland, K. N. (2013). Tinbergen's four questions: An appreciation and an update. *Trends in Ecology & Evolution, 28*, 712–718.

Engelhardt, S. C., & Taborsky, M. (2023). Reciprocal altruism in Norway rats. *Ethology, 130*, e13418.

Fraser, H., Parker, T., Nakagawa, S., Barnett, A., & Fidler, F. (2018). Questionable research practices in ecology and evolution. *PLOS ONE, 13*, e0200303.

Green, K. K., & Madjidian, J. A. (2011). Active males, reactive females: Stereotypic sex roles in sexual conflict research? *Animal Behaviour, 81*, 901–907.

Kamath, A., & Wesner, A. B. (2020). Animal territoriality, property and access: A collaborative exchange between animal behaviour and the social sciences. *Animal Behaviour, 164*, 233–239.

Ligon, R. A., & McGraw, K. J. (2016). Social costs enforce honesty of a dynamic signal of motivation. *Proceedings of the Royal Society B: Biological Sciences, 283*, 20161873.

Nesse, R. M. (2013). Tinbergen's four questions, organized: A response to Bateson and Laland. *Trends in Ecology & Evolution, 28*, 681–683.

Panware, P., Gómez-Ruiz, P. A., Zipple, M. N., & Sandoval, L. (2022). Changes in escape behaviour of black iguanas (*Ctenosaura similis*) over an anthropogenic disturbance gradient. *Behaviour, 159*, 111–131.

Richardson, J., & Zuk, M. (2023). Unlike a virgin: A meta-analytical review of female mating status in studies of female choice. *Behavioral Ecology, 34*, 165–182.

Scordato, E. S. C. (2018). Male competition drives song divergence along an ecological gradient in an avian ring species. *Evolution, 72*, 2360–2377.

Suárez-Rodríguez, M., López-Rull, I., & Garcia, C. M. (2012). Incorporation of cigarette butts into nests reduces nest ectoparasite load in urban birds: New ingredients for an old recipe? *Biology Letters, 9*, 20120931.

Assignments and Assessments

1. **Participation** (10%, Aligned to LO1-LO5)

   Active participation is expected throughout the course and contributes 10% to the final grade. Participation is divided into two components:

   Article Discussions (5%): Engage thoughtfully in discussions of assigned readings, demonstrating understanding of key concepts. Contributions should be substantive, respectful, and show evidence of preparation.

   General Participation (5%): Contribute to class activities, discussions, and group work. This includes asking questions, offering insights, and collaborating effectively with peers. Consistent engagement throughout the course will be reflected in this portion of the grade.

2. **Lecture Exams** (30%, Aligned to LO1-LO5)

   There will be three online asynchronous exams, each worth 10% of your final grade. Exams will include a mix of short-answer and open-ended questions, designed to test your understanding of key concepts and your ability to apply what you've learned. Be sure to review lecture material, notes, and readings to prepare.

   Exam 1 (10%): Evolution & behavior basics
   Exam 2 (10%): Predation/foraging, habitat use, communication
   Exam 3 (10%): Mating/reproduction, parental care, sociality

3. **Paired Problem-Solving** (15%, Aligned to LO1-LO3)

   To practice the critical skills of behavior research, we will be engaging in paired-partner activities throughout the course. These activities will take place during class, and will include filling in an activity sheet which will be submitted online and graded at the conclusion of the activity:

   Video scoring (5%): Paired data logging and analysis of recorded animal behaviors
   Experimental design challenge (5%): Design & experimental confound problem solving
   Data detectives (5%): Data analysis and drawing conclusions from results

4. **Research Project** (35%, Aligned to LO1-LO5)
   Students will complete a course-long individual research project to investigate specific behaviors of animals in their local community. The design and completion of this project is broken up into four major steps, each of which offers opportunities for feedback and guidance:

   Ethogram and Presentation (5%): An ethogram is a comprehensive list of behaviors observed in one or more individuals of a single species. The purpose of the ethogram is to build a catalog of the general "behavioral repertoire" of the species of interest. This assignment includes both turning in a written ethogram (2.5%) and preparing a 2-3 minute in-class presentation of the written report (2.5%).

   Structured Observations and Presentation (5%): Using their ethograms, students will observe and collect data on the behaviors of focal individuals of their species of interest. The purpose of this activity is to begin developing a set of questions which can be addressed through observation or experimentation. This assignment includes both turning in written observations (3%) and a 2-3 minute in-class presentation of the written report (2%)

   Proposal (10%): Before collecting data for the final report, students will submit a project proposal. The preliminary proposal (2.5%) will be graded on completion, and the final proposal (7.5%) will be graded based on the rigor of the proposal, and how well the students respond to the suggestions of faculty and peers.

   Final report (15%): A comprehensive report on the question, methods, findings, and implications of the research project.

5. **Presentation and Peer Reviews** (10%, Aligned to LO3-LO5)

We will be practicing critical analysis and peer review of each others' projects. Students will be responsible for watching presentations (in groups of 4-6) and providing feedback using a standardized submission format. Each student will also submit peer reviews for two classmates based on a thorough reading and assessment of their written proposals. Comprehensive and thoughtful comments on the following components of classmates' research projects will be graded:

Ethogram (2.5%) - Comments submitted after in-class presentations
Structured Observations (2.5%) - Comments submitted after in-class presentations
Proposal Peer Review (5%) - Written feedback submitted as a homework assignment

Grading

The final grade will be calculated as described below:

Assignment/Assessment	% Weight	Individual or Group/Team Grade
Participation	10%	Individual
Lecture Exams	30%	Individual
Paired Problem-Solving	15%	Group
Presentation and Peer Reviews	10%	Individual
Research Project	35%	Individual

Course Schedule/Course Calendar

Module/Week

Module with specific dates	Topic	Readings	Activities/Assignments for this module
Module 1 Monday July 6	What is behavior? Phenotypes, heritability, & evolution	Required: ● **Suárez-Rodríguez M, López-Rull I,Garcia CM.** 2012. Incorporation of cigarette butts into nests reduces nest ectoparasite load in urban birds: new ingredients for an old recipe? Biology Letters, 9:20120931. ● **Rubenstein &Alcock**: Chapter1	ASSIGNMENTS ● Participate in "Getting to Know You Discussion" forum before the week begins IN CLASS ● **Activity:** What is behavior? ● **Discussion:** Suárez-Rodríguez *et al.*

Module 2 Wednesday July 8	Natural selection, dynamics, constraints, plasticity Behavioral toolkit: sampling & observation	Required: • **Altmann J.** 1974. Observational study of behavior: sampling methods. Behaviour, 49:227-266. • **Bateson P, Laland KN**. 2013. Tinbergen's four questions:an appreciation and an update.Trends in Ecology & Evolution, 28: 712-718. • **Nesse RM**. 2013. Tinbergen's four questions, organized: a response to Bateson and Laland. Trends in Ecology & Evolution, 28: 681-683	IN CLASS • **Discussion:** Bateson & Laland, Nesse • **Activity:** Trying out Tinbergen's 4 questions
Module 3 Monday July 13	Predation & foraging Behavioral toolkit: experimental design	Required: • **Panware P, Gómez-Ruiz PA, Zipple MN, Sandoval L**. 2022. Changes in escape behaviour of black iguanas (Ctenosaura similis) over an anthropogenic disturbance gradient. Behaviour, 159: 111-131. • **Rubenstein & Alcock**: Chapter 6 (pp. 184 – 217)	ASSIGNMENTS • **Research Project:** Ethogram observations IN CLASS • **Discussion**: Panware et al. • **Research Project:** Ethogram presentations
Module 4 Wednesday July 15	Niches, habitat use, & territoriality	Required: • **Kamath A, Wesner AB**. 2020. Animal territoriality, property and access: a collaborative exchange between animal behaviour and the social sciences. Animal Behaviour, 164: 233-239. • **Rubenstein & Alcock**: Chapter 7 (pp. 220 – 235)	ASSIGNMENTS • **EXAM 1:** Evolution & Behavior Basics IN CLASS • **Paired problem-solving**: Video scoring (submit worksheet) • **Discussion:** Kamath & Wesner
Module 5 Monday July 20	Communication: signals and cues	Required: • **Rubenstein & Alcock**: Chapter 8 (pp. 258 – 262)	ASSIGNMENTS • **Research Project:** Structured observations IN CLASS

			• **Research Project:** Structured observation presentations
Module 6 Wednesday July 22	Honesty & deceit, the origins of signals	Required: • **Ligon RA, McGraw KJ**. 2016. Social costs enforce honesty of a dynamic signal of motivation. Proceedings of the Royal Society B, 283: 20161873. • **Rubenstein & Alcock**: Chapter 8 (pp. 262 – 295)	ASSIGNMENTS • **Research Project:** Preliminary project proposal IN CLASS • **Discussion**: Ligon & McGraw • **Paired problem-solving:** Experimental design challenge (submit worksheet)
Module 7 Monday July 27	Reproductive behavior & mating systems	Required: • **Richardson J, Zuk M**. 2023. Unlike a virgin: a meta-analytical review of female mating status in studies of female choice. Behavioral Ecology, 34: 165-182. • **Rubenstein & Alcock**: Chapter 10 (pp. 355 – 397)	ASSIGNMENTS • **Research Project:** Proposal peer reviews • **EXAM 2**: Predation/foraging, Habitat use, Communication IN CLASS • **Discussion:** Richardson & Zuk
Module 8 Wednesday July 29	Sexual selection: mate choice and competition	Required: • **Scordato ESC. 2018**. Male competition drives song divergence along an ecological gradient in an avian ring species. Evolution, 72: 2360-2377. • **Rubenstein & Alcock**: Chapter 9 (pp. 305 – 343)	ASSIGNMENTS • **Research Project:** Final project proposal IN CLASS • **Discussion:** Sex, gender, and the study of sexual behaviors • **Discussion:** Scordato

Module 9 Monday Aug 3	Parental care	Required: ● **Rubenstein & Alcock**: Chapter 11	ASSIGNMENTS ● Continue collecting project data IN CLASS ● **Paired problem-solving**: Data Detectives (submit worksheet) ● **Discussion/activity:** Crafting figures & tables
Module 10 Wednesday Aug 5	Social behavior, altruism, & eusociality	Required: ● **Engelhardt SC, Taborsky M.** 2023. Reciprocal altruism in Norway rats. Ethology, 2024:130e13418. ● **Rubenstein & Alcock**: Chapters 12 & 13	ASSIGNMENTS ● Continue collecting project data IN CLASS ● **Discussion**: Engelhardt & Taborsky ● **Discussion/activity:** Crafting a scientific argument
Module 11 Monday Aug 10	Ethics, bias, and humanity in the study of animals	Required: ● **Fraser H, Parker T, Nakagawa S, Barnett A, Fidler F.** 2018. Questionable research practices in ecology and evolution. PLOS One, 13: e0200303. ● **Green KK, Madjidian JA.** 2011. Active males, reactive females: stereotypic sex roles in sexual conflict research? Animal Behaviour, 81: 901-907.	ASSIGNMENTS ● **EXAM 3:** Mating, reproduction, Parental care, sociality IN CLASS ● **Discussion:** Fraser et al.; Green & Madjidian
Module 12 Friday Aug 14			ASSIGNMENTS ● **Research Project:** Final papers due

Course Policies

Participation

Because of the compressed summer timeline, this is going to be a fast-moving course. Our seminar time is surprisingly brief, and a successful semester depends on your attendance and engagement. Missing a class or turning in an assignment late will make it difficult for you to keep up with our scheduled activities.

Students are permitted one unexcused absence, but further absences will adversely affect your final grade. If you are prevented from attending due to an unexpected emergency (i.e., illness, injury, other personal issues), please notify me as soon as possible and provide relevant documentation. If you expect to be absent for a non-emergency reason (e.g., travel for sporting events or conferences, planned attendance at a conflicting campus event, etc.) please let me know as soon as possible. **I will not retrospectively excuse absences due to non-emergency events** (i.e., planned trips, sporting events, conflicting schedules with other classes).

Late work

All assignments must be turned in by their specified deadlines or your grade will be penalized. We will be sharing and discussing our work in class, so missing deadlines will also affect your ability to fully participate.

Late submissions will be reduced by one letter grade per calendar day late. As with attendance, I will be happy to extend a homework deadline due to an unexpected emergency given the conditions outlined above. I am also happy to discuss extensions for non-emergency reasons, but **I will not retrospectively offer extensions due to non-emergency events.**

Citation & Submission

All written assignments must cite all relevant sources using APA citation formatting, and must be submitted to the course website (not via an email to the instructor).

Statement on AI in the classroom

This course is intended to build critical thinking skills and rigorous scientific methods of problem solving. This includes digesting information, explaining complex topics, and generating novel insights in both classroom discussions and writing. We will focus on activities which:

1. **Show thinking:** e.g. take notes on journal articles by hand, write discussion questions in collaboration with faculty
2. **Draw on class conversations:** e.g., write reports based on observations made by peers in discussion, provide written or verbal feedback to presentations by students in class groups
3. **Emphasize data and observation:** e.g., share findings with peers, present experiment proposals, focus on opportunities to present data and ideas over formatting and presentation style

The use of generative AI (i.e., ChatGPT and other LLMs) is antithetical to these activities, and students are expected to read and write their own materials. As with any actions that outsource your thinking or writing to others, the unauthorized use of AI is considered academic dishonesty and will result in a failing grade.

Academic Honesty

Behaving with integrity is essential to the value of your Columbia education and degree. You are here to develop your skills and knowledge, and the grade you receive in this course is intended to reflect your mastery of the content. You must behave in accordance with instructions for all assignments and assessments, by using only the

allowed resources and tools and by not sharing your answers and work when such collaboration is explicitly disallowed. Violation of the rules of academic integrity will result in an automatic failing grade for the course. For more information, please visit college.columbia.edu/academics/academicintegrity

Disability Accommodations

If you wish to receive disability-related academic accommodations, make sure to register with Disability Services. Faculty must be notified of registered students' accommodations before exams or other accommodations will be provided. If you have a disability, or would like to confidentially discuss potential accommodations, please contact Columbia ODS (email: disability@columbia.edu, tel: 212-854-2388) or Barnard CARDS (email: cards@barnard.edu, tel: 212-854-4634).

School and University Policies and Resources

Copyright Policy
Please note—Due to copyright restrictions, online access to this material is limited to instructors and students currently registered for this course. Please be advised that by clicking the link to the electronic materials in this course, you have read and accept the following:

> The copyright law of the United States (Title 17, United States Code) governs the making of photocopies or other reproductions of copyrighted materials. Under certain conditions specified in the law, libraries and archives are authorized to furnish a photocopy or other reproduction. One of these specified conditions is that the photocopy or reproduction is not to be "used for any purpose other than private study, scholarship, or research." If a user makes a request for, or later uses, a photocopy or reproduction for purposes in excess of "fair use," that user may be liable for copyright infringement.

Academic Integrity
Columbia University expects its students to act with honesty and propriety at all times and to respect the rights of others. It is fundamental University policy that academic dishonesty in any guise or personal conduct of any sort that disrupts the life of the University or denigrates or endangers members of the University community is unacceptable and will be dealt with severely. It is essential to the academic integrity and vitality of this community that individuals do their own work and properly acknowledge the circumstances, ideas, sources, and assistance upon which that work is based. Academic honesty in class assignments and exams is expected of all students at all times.

Civility Statement
It is our intent that students from all backgrounds and perspectives be well-served by this course, that students' learning needs be addressed both in and out of class, and that the diverse perspectives that the students bring to this class be viewed as a resource, strength, and benefit. It is our intent to present materials and activities that are respectful of all participants.

Accessibility
Columbia is committed to providing equal access to qualified students with documented disabilities. A student's disability status and reasonable accommodations are individually determined based upon disability documentation and related information gathered through the intake process. For more information regarding this service, please visit the University's Health Services website: https://health.columbia.edu/content/disability-services.

Class Recordings
All or portions of the class may be recorded at the discretion of the Instructor to support your learning. At any point, the Instructor has the right to discontinue the recording if it is deemed to be obstructive to the learning process.

If the recording is posted, it is confidential and it is prohibited to share the recording outside of the class.

Columbia University Information Technology
Columbia University Information Technology (CUIT) provides Columbia University students, faculty and staff with central computing and communications services. Students, faculty and staff may access University-provided and discounted software downloads.

Columbia University Library
Columbia's extensive library system ranks in the top five academic libraries in the nation, with many of its services and resources available online.

Netiquette
Online sessions in this course will be offered through Zoom, accessible through Canvas. A reliable Internet connection and functioning webcam and microphone are required. It is your responsibility to resolve any known technical issues prior to class. Your webcam should remain turned on for the duration of each class, and you should expect to be present the entire time. Avoid distractions and maintain professional etiquette.

Please note: Instructors may use Canvas or Zoom analytics in evaluating your online participation.

More guidance can be found at: https://jolt.merlot.org/vol6no1/mintu-wimsatt_0310.htm

Netiquette is a way of defining professionalism for collaborations and communication that take place in online environments. Here are some Student Guidelines for this class:

- Avoid using offensive language or language that is not appropriate for a professional setting.
- Do not criticize or mock someone's abilities or skills.
- Communicate in a way that is clear, accurate and easy for others to understand.
- Balance collegiality with academic honesty.
- Keep an open-mind and be willing to express your opinion.
- Reflect on your statements and how they might impact others.
- Do not hesitate to ask for feedback.
- When in doubt, always check with your instructor for clarification.